TIME-OF-FLIGHT DISTANCE SENSOR

# ARK DIST

DroneCAN and MAVLink laser distance sensor in short-, mid- and long-range variants

EXPOSED INTERFACES CAN UART SWD

The USA-built ARK DIST pairs a Broadcom AFBR-S50 time-of-flight sensor with an STM32F412VG running the PX4 DroneCAN node firmware. The SR, MR and LR variants share one board and case and differ only in the sensor module.



## 01 KEY FEATURES

- Broadcom AFBR-S50 time-of-flight sensor, 850 nm Class 1 laser
- Max reported distance: SR 30 m, MR 50 m, LR 100 m
- DroneCAN and MAVLink 2 output, firmware update over CAN
- 2× CAN and UART on Pixhawk-standard JST-GH, debug on JST-SH
- PX4 and ArduPilot; ROS 2 via ros2_dronecan

## 02 SPECIFICATIONS

| SENSOR | |
|---|---|
| Time-of-flight module | Broadcom AFBR-S50, see 03 |
| Laser | 850 nm, IEC 60825-1 Class 1 (eye-safe) |
| Transmitter beam | 2° × 2°, 3.5 cm spot at 1 m |
| Receiver (SR, MR) | 32 pixels, 12.4° × 6.2° field of view |

| INTERFACES | |
|---|---|
| DroneCAN | uavcan.equipment.range_sensor.Measurement |
| CAN functions | Firmware update, dynamic node allocation |
| UART | MAVLink 2 DISTANCE_SENSOR and HEARTBEAT |
| UART settings | 115200 8N1, RTS/CTS optional (auto-detected) |
| MAVLink system / component ID | 158 / 158 |
| Debug | NSH console and SWD |

| PROCESSOR & FIRMWARE | |
|---|---|
| MCU | STM32F412VG |
| Firmware target | ark_dist_default, board ID 92 |
| First release with all variants | ark_dist-v1.18.0.1 |

| ELECTRICAL & MECHANICAL | |
|---|---|
| Supply voltage | 5 V |
| Supply current | 78–86 mA |
| Dimensions | 20 × 28 × 14 mm |
| Weight | 4 g |

| AT A GLANCE | |
|---|---|
| Sensor | Broadcom AFBR-S50 |
| Variants | SR / MR / LR |
| Max reported distance | 30 / 50 / 100 m |
| Update rate | 20 / 15 / 10 Hz |
| Supply | 5 V |
| Dimensions | 20 × 28 × 14 mm |
| Weight | 4 g |
| Hardware rev | Rev 1.0 |

### COMPLIANCE & SOURCING

- Made in the USA
- NDAA compliant
- IEC 60825-1 Class 1 laser product

### ORDERING & RESOURCES

**STORE** arkelectron.com/product/ark-dist-sr/
arkelectron.com/product/ark-dist-mr/
arkelectron.com/product/ark-dist-lr/

**DOCS** docs.arkelectron.com/products/sensor/ark-dist

**CAD** github.com/ARK-Electronics/ark-hardware

## 03 VARIANTS

| | ARK DIST SR | ARK DIST MR | ARK DIST LR |
|---|---|---|---|
| Time-of-flight module | AFBR-S50LV85D | AFBR-S50LX85D | AFBR-S50EX85D |
| Measurement mode | Long Range, DFM 4X | Long Range, DFM 4X | Extended Long Range |
| Update rate | 20 Hz (Fast profile 50 Hz) | 15 Hz (Fast profile 25 Hz) | 10 Hz |
| Max reported distance | 30 m | 50 m | 100 m |
| Min reported distance | 0 m | 0.1 m | 0.6 m |

Mode, rate and reported-distance limits are shipped firmware defaults. The Fast profile is SENS_AFBR_PROF = 1.

## 04 RANGE

| SURFACE | ARK DIST SR | | ARK DIST MR | | ARK DIST LR | |
|---|---|---|---|---|---|---|
| | BRIGHT SUN | DUSK | BRIGHT SUN | DUSK | BRIGHT SUN | DUSK |
| Asphalt | 18 m | 30 m | 18 m | 40 m | 18 m | 40 m |
| Grass | 18 m | 30 m* | 18 m | 50 m* | 18 m | 82 m |

Reliable range: the farthest distance with ≥50 % of readings valid and within 5 m of the reference, at default settings. Bright sun: sun 49–61° above the horizon; a lower sun gives more range. Dusk: sun below 6°. * Limited by the maximum reported distance; the sensor was still measuring at the flight ceiling (SR 38 m, MR 54 m). Wet surfaces reduce range; open water (~2 % reflectivity) gives only short, intermittent returns.

## 05 CAN CONNECTOR (×2)

| PIN | SIGNAL | VOLTAGE |
|---|---|---|
| 1 | 5V | 5.0 V |
| 2 | CAN_P | 5.0 V |
| 3 | CAN_N | 5.0 V |
| 4 | GND | GND |

Connector: 4-pin JST-GH, Pixhawk standard. Chain additional nodes from the second CAN connector.

## 06 UART CONNECTOR

| PIN | SIGNAL | VOLTAGE |
|---|---|---|
| 1 | 5V_UART | 5.0 V |
| 2 | USART3_RX | 3.3 V |
| 3 | USART3_TX | 3.3 V |
| 4 | USART3_RTS | 3.3 V |
| 5 | USART3_CTS | 3.3 V |
| 6 | GND | GND |

Connector: 6-pin JST-GH, Pixhawk standard.

## 07 DEBUG CONNECTOR

| PIN | SIGNAL | VOLTAGE |
|---|---|---|
| 1 | 3.3V | 3.3 V |
| 2 | USART2_TX | 3.3 V |
| 3 | USART2_RX | 3.3 V |
| 4 | FMU_SWDIO | 3.3 V |
| 5 | FMU_SWCLK | 3.3 V |
| 6 | GND | GND |

Connector: 6-pin JST-SH, Pixhawk standard.

## 08 LED INDICATORS

| STATUS LED | MEANING |
|---|---|
| Green, fast blink (10 Hz) | Node firmware running. It does not change with CAN traffic. |
| Green, slow blink (1 Hz) | Bootloader listening for CAN traffic |
| Blue, blinking (2 Hz) | Bootloader waiting for a node ID from the flight controller |
| Blue and green, blinking (3 Hz) | Bootloader waiting up to 3 s for the flight controller to start a firmware update |
| Red, green and blue together (3 Hz) | Firmware update in progress |
| Red, blinking | Firmware update error. 1 Hz: the flight controller returned a file error. 2 Hz: the flight controller stopped responding. 4 Hz: the image failed its CRC check. A failed update restarts the node after 20 s. |

Without valid firmware the node stays in the bootloader until the flight controller flashes it.

ARK-DS-ARK-DIST | Issue 2026-10-12 | Compiled from ARK Electronics published documentation, the ark_dist-v1.18.0.1 firmware defaults, and ARK flight measurements. Laser and optical characteristics per Broadcom. Specifications are subject to change without notice. © 2026 ARK Electronics LLC.